

CSE 5825 – Bayesian Machine Learning

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Bayesian machine learning is a unifying methodology for reasoning about uncertainty when modelling complex data. This course begins by covering the foundations of probabilistic modelling, Monte Carlo and variational inference algorithms, and model checking. We build on these foundations by considering essential models, e.g., mixed-membership and hierarchical models, and their applications. The course concludes with a survey of recent advances in Bayesian machine learning focusing on Bayesian nonparametrics and other advanced topics. We will reinforce class lectures by developing a research project (ideally connected to your research area) on experimental data that implements this probabilistic framework from problem specification to model checking. To realize this, you should be comfortable with (or willing to learn):

- basic probability, statistics, and calculus
- writing software to analyze large data and some statistical supportive programming language like Python or R
- scientific writing

1 Logistics

The course instructor is Derek Aguiar (derek.aguiar@uconn.edu); office is ITE 267.

Class times are Tuesday and Thursday at 2:00pm - 3:15pm in ITE 127.

Collaboration hours are on Wed. 5:30pm-7pm and Thursday 3:30pm - 5pm in ITE 230 (take CSE main elevator to 2nd floor, go right to the far corridor, then right again).

The course website is hosted on HuskyCT which is used for course material distribution, announcements, lectures. We also use Perusall for collaboratively reading course materials.

Discussions outside of course will take place on Discord, (see HuskyCT).

Please hand in homeworks on time, but, if you need extra time due to some hardship, don't be shy about asking for more time (email me).

Please fill out the course questionnaire at <https://forms.office.com/r/CC9BtNuXwh>

1.1 Course requirements

- Required reading (using Perusall) – **10%**
- 2 homework assignments (each requires multiple weeks) – **40%**
- An original research project **or** an extensive review on a subject relevant to the course. This includes project progress reports, a final deliverable, and the final presentation at the end of the semester. – **50%**

We encourage LaTeX or Markdown for typesetting work. You must show contributions to Perusall to receive the 10% credit for the reading. At least 1 Perusall contribution per week is required for full credit.

2 Schedule

A tentative schedule of topics and sample readings (not necessarily required) are included below. A current schedule with readings will be available on the website.

1. Graphical and statistical models

- David Barber. *Bayesian reasoning and machine learning*. Cambridge University Press, 2012. URL: <http://web4.cs.ucl.ac.uk/staff/D.Barber/textbook/090310.pdf>
- Michael I Jordan. *An introduction to probabilistic graphical models*. 2003. URL: <https://people.eecs.berkeley.edu/~jordan/prelims/>
- Kevin Murphy. “An introduction to graphical models”. In: *Rap. tech* (2001), pp. 1–19
- Daphne Koller and Nir Friedman. *Probabilistic graphical models: principles and techniques*. MIT press, 2009

2. Inference

- Radford M Neal. “Probabilistic inference using Markov chain Monte Carlo methods”. In: (1993)
- Philip Resnik and Eric Hardisty. *Gibbs sampling for the uninitiated*. Tech. rep. 2010
- David M Blei, Alp Kucukelbir, and Jon D McAuliffe. “Variational inference: A review for statisticians”. In: *Journal of the American Statistical Association* 112.518 (2017), pp. 859–877

3. Models

- David M Blei, Andrew Y Ng, and Michael I Jordan. “Latent dirichlet allocation”. In: *Journal of machine Learning research* 3.Jan (2003), pp. 993–1022
- Jonathan K Pritchard, Matthew Stephens, and Peter Donnelly. “Inference of population structure using multilocus genotype data”. In: *Genetics* 155.2 (2000), pp. 945–959

4. Posterior Checking

- Andrew Gelman, Xiao-Li Meng, and Hal Stern. “Posterior predictive assessment of model fitness via realized discrepancies”. In: *Statistica sinica* (1996), pp. 733–760
- Andrew Gelman. “A Bayesian formulation of exploratory data analysis and goodness-of-fit testing”. In: *International Statistical Review* 71.2 (2003), pp. 369–382

5. Implementation

- Dustin Tran et al. “Edward: A library for probabilistic modeling, inference, and criticism”. In: *arXiv preprint arXiv:1610.09787* (2016)
- Bob Carpenter et al. “Stan: A probabilistic programming language”. In: *Journal of statistical software* 76.1 (2017)

6. Bayesian nonparametrics

- Samuel J Gershman and David M Blei. “A tutorial on Bayesian nonparametric models”. In: *Journal of Mathematical Psychology* 56.1 (2012), pp. 1–12
- Yee W Teh et al. “Sharing clusters among related groups: Hierarchical Dirichlet processes”. In: *Advances in neural information processing systems*. 2005, pp. 1385–1392

7. Representation Learning and Deep generative models

- Tschannen, Michael, Olivier Bachem, and Mario Lucic. ”Recent advances in autoencoder-based representation learning.” *arXiv preprint arXiv:1812.05069* (2018).
- Cao, Hanqun, et al. ”A survey on generative diffusion models.” *IEEE Transactions on Knowledge and Data Engineering* (2024).

3 Resources

3.1 Textbook

There is no required textbook for the course, although, there is assigned required reading, optional reading, and implicit reading required for your final projects. Readings will mostly come research articles; they will be linked to from the website. Another potentially useful (and free) resources is David Barber. *Bayesian reasoning and machine learning*. Cambridge University Press, 2012. URL: <http://web4.cs.ucl.ac.uk/staff/D.Barber/textbook/090310.pdf>;

The following are also good references for various topics covered in this course.

- Christopher M Bishop. *Pattern recognition and machine learning*. Tech. rep. 2006
- Kevin P. Murphy. *Probabilistic Machine Learning: An introduction*. MIT Press, 2022. URL: probml.ai and Kevin P. Murphy. *Probabilistic Machine Learning: Advanced Topics*. MIT Press, 2023. URL: <http://probml.github.io/book2> or the previous version which is better in many ways: Kevin P Murphy. *Machine Learning: A Probabilistic Perspective*. MIT Press, 2012
- Andrew Gelman et al. *Bayesian data analysis*. Chapman and Hall/CRC, 1995
- Daphne Koller and Nir Friedman. *Probabilistic graphical models: principles and techniques*. MIT press, 2009

3.2 Collaboration and Using AI

For any assignment, we strongly encourage high-level discussion of ideas and problems with living things – e.g., the instructor, classmates, cats, dogs, or houseplants. If there is ambiguity about what constitutes “high-level discussion”, the category of the assignment, or anything else regarding assignments, please ask the instructor. Collaboration hours and the course Discord are excellent places to ask questions and discuss ideas. Students may also use AI tools to support learning. However, keep in mind that using AI tools has the potential to undermine your learning. Appropriate use includes requesting hints, conceptual clarification, or guidance on small parts of a problem. AI should not be used to generate full solutions, proofs, or completed code. The goal is for AI to help you reach the “A-HA!” moment, not replace it. All submitted work must reflect your own understanding.

3.3 Instructor use of AI

The instructor has not used AI to design the course and will not use AI to grade student work. Instructor use of AI is limited to rephrasing instructor-generated text and generating supplementary content that supports student learning, such as sample problems.

3.4 Software

The ability to reproduce results and extend or reuse existing code are essential to research. We highly recommend using Github or some other version control system to document your progress on your final project.

3.5 Writing help

Learning how to write a scientific article is an arduous process. At UConn, the [writing center](#) can be a great resource. Other good resources include:

- Paul J Silvia. *How to write a lot: A practical guide to productive academic writing*. American Psychological Association, 2007
- Wendy Laura Belcher. *Writing your journal article in twelve weeks: A guide to academic publishing success*. Sage, 2009

References

- Barber, David. *Bayesian reasoning and machine learning*. Cambridge University Press, 2012. URL: <http://web4.cs.ucl.ac.uk/staff/D.Barber/textbook/090310.pdf>.
- Belcher, Wendy Laura. *Writing your journal article in twelve weeks: A guide to academic publishing success*. Sage, 2009.
- Bishop, Christopher M. *Pattern recognition and machine learning*. Tech. rep. 2006.
- Blei, David M, Alp Kucukelbir, and Jon D McAuliffe. “Variational inference: A review for statisticians”. In: *Journal of the American Statistical Association* 112.518 (2017), pp. 859–877.
- Blei, David M, Andrew Y Ng, and Michael I Jordan. “Latent dirichlet allocation”. In: *Journal of machine Learning research* 3.Jan (2003), pp. 993–1022.
- Carpenter, Bob et al. “Stan: A probabilistic programming language”. In: *Journal of statistical software* 76.1 (2017).
- Gelman, Andrew. “A Bayesian formulation of exploratory data analysis and goodness-of-fit testing”. In: *International Statistical Review* 71.2 (2003), pp. 369–382.
- Gelman, Andrew, Xiao-Li Meng, and Hal Stern. “Posterior predictive assessment of model fitness via realized discrepancies”. In: *Statistica sinica* (1996), pp. 733–760.
- Gelman, Andrew et al. *Bayesian data analysis*. Chapman and Hall/CRC, 1995.
- Gershman, Samuel J and David M Blei. “A tutorial on Bayesian nonparametric models”. In: *Journal of Mathematical Psychology* 56.1 (2012), pp. 1–12.
- Jordan, Michael I. *An introduction to probabilistic graphical models*. 2003. URL: <https://people.eecs.berkeley.edu/~jordan/prelims/>.
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- Murphy, Kevin. “An introduction to graphical models”. In: *Rap. tech* (2001), pp. 1–19.
- Murphy, Kevin P. *Machine Learning: A Probabilistic Perspective*. MIT Press, 2012.
- Murphy, Kevin P. *Probabilistic Machine Learning: Advanced Topics*. MIT Press, 2023. URL: <http://probml.github.io/book2>.
- *Probabilistic Machine Learning: An introduction*. MIT Press, 2022. URL: probml.ai.
- Neal, Radford M. “Probabilistic inference using Markov chain Monte Carlo methods”. In: (1993).
- Pritchard, Jonathan K, Matthew Stephens, and Peter Donnelly. “Inference of population structure using multilocus genotype data”. In: *Genetics* 155.2 (2000), pp. 945–959.
- Resnik, Philip and Eric Hardisty. *Gibbs sampling for the uninitiated*. Tech. rep. 2010.
- Silvia, Paul J. *How to write a lot: A practical guide to productive academic writing*. American Psychological Association, 2007.
- Teh, Yee W et al. “Sharing clusters among related groups: Hierarchical Dirichlet processes”. In: *Advances in neural information processing systems*. 2005, pp. 1385–1392.
- Tran, Dustin et al. “Edward: A library for probabilistic modeling, inference, and criticism”. In: *arXiv preprint arXiv:1610.09787* (2016).